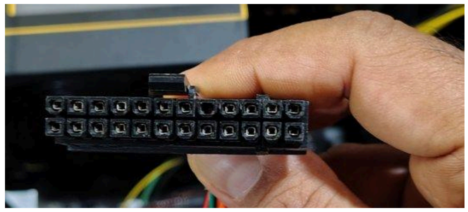
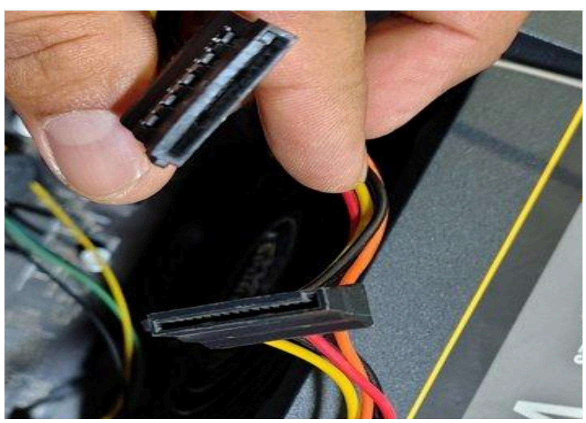
Task 3 - Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

1. ATX 24-pin motherboard connector



Name	ATX 24-pin main power connector
Devices powered	Connects directly to the motherboard and supplies the +3.3V, +5V, +12V, and standby power rails that run the chipset, RAM, and other onboard circuitry.
Safety tip	Always switch off and unplug the SMPS from the wall before connecting or disconnecting this connector, and make sure the locking clip clicks fully into place so it cannot work loose.

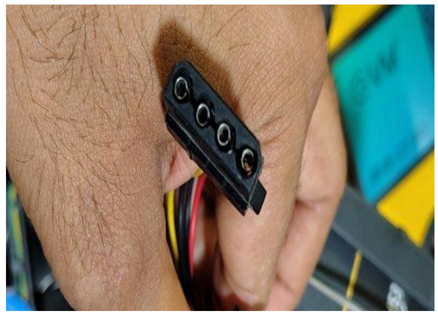
2. SATA power connector



Name	SATA power connector (flat, L-shaped, with orange/red/yellow/black wires)
Devices powered	Powers SATA-interface storage devices such as SSDs and hard disk drives, supplying 3.3V (orange), 5V (red), and 12V (yellow) rails.

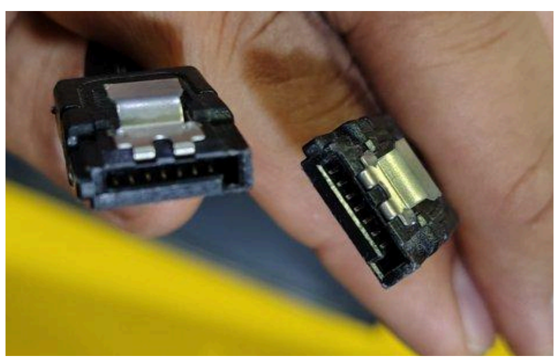
Safety tip	Insert it gently in the correct orientation until it clicks — the connector is keyed so it only fits one way, and forcing it in upside down can damage the drive's power pins.

3. Molex (4-pin peripheral) connector



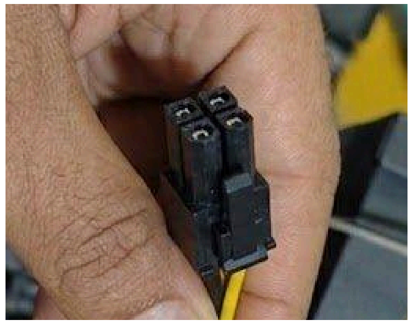
Name	Molex 4-pin connector
Devices powered	Used for older optical drives, case fans, and some fan/RGB controllers; carries +5V and +12V.
Safety tip	Check that the bevelled (chamfered) corners line up before pushing it in — it can be forced in the wrong way round, which risks bending pins or reversing polarity.

4. SATA data connector —



Name	SATA data connector (7-pin)
Devices powered	Carries data between the motherboard's SATA port and the drive (SSD/HDD) — this comes from the motherboard, not the PSU
Safety tip	Insert gently along the keyed notch; avoid bending the thin data pins, which are more fragile than the power pins

CPU ATX 4-pin 12V (EPS) connector



Name	CPU ATX 4-pin 12V connector (EPS)
Devices powered	Plugs into the dedicated 4-pin socket near the CPU socket on the motherboard, supplying focused +12V power just for the processor (separate from the main 24-pin rail).
Safety tip	Make sure the small locking clip on the connector engages fully — a loose CPU power connection can cause random shutdowns or failure to boot under load.